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Customer Information

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Performance, accuracy and repeatability of new Endress+Hauser pH and ORP sensors

The best performance of a pH measurement can be achieved by use of traceable high quality pH buffer solutions for calibration and adjustment. For highly accurate and reproducible pH measurements, the pH sensor must have a slope of at least 58 mV/pH, the asymmetry should not be bigger than 10 mV with glass electrodes, the operational point of ISFET sensors not bigger than ± 5 mV.

Under those conditions liquid KCl (e.g. CPS41D) and gel filled sensors (e.g. CPS11D, CPS71D, CPSx7D, CPF81D, CPS171D) can reach under best conditions in an accredited laboratory an accuracy of ± 0.028 pH with a repeatability of ± 0.01 pH. They withstand a slight vacuum of 0.8 bar abs, e.g. after sterilization in place (SIP) or cleaning in place (CIP).

The ORP electrodes (e.g. CPS12D, CPS42D, CPS72D, CPF82D) can reach an accuracy of ± 5 mV. That is due to the fact, that the commercially available ORP buffer solutions have a best accuracy of ± 5 mV. There is no traceable standard reference material available.

Under controlled and constant temperature conditions (± 0.5 K) the above mentioned ORP electrodes can reach a reproducibility of ± 3 mV. ORP is a sum parameter and can't be temperature compensated.



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